Dyno Nobel Limited (ASX:DNL)

Scenario-based valuation briefing

Prepared for: **Meeting with Ben** | Date: 29 May 2026 | Author: Tara Reid, Reid Advisory

Headline

Post-demerger Dyno Nobel is a pure-play industrial-explosives company with a structurally defensible position — co-leader globally, duopolist in Australia, US cost-curve advantage from long-term gas contracts — but with material exposure to cyclical mining-customer demand, energy and ammonia input costs, and a coal-customer mix that is structurally declining over the long horizon. Across the six scenarios, DNL is negatively asymmetric — downside scenarios (Stagflation, Disorderly Climate, Fragmentation) compress value materially more than the upside (Orderly Convergence) lifts it. Single-WACC discipline confirms the asymmetry sits in the cash-flow path, not in discount-rate gymnastics.

Per-scenario value summary

Scenario	Per share	vs market	vs Muddle Through
Orderly Convergence	AUD 3.72	+3%	+16%
Muddle Through (central)	**AUD 3.22**	**-11%**	**baseline**
AI Productivity Lag	AUD 3.14	-13%	-2%
Fragmentation	AUD 2.36	-35%	-27%
Disorderly Climate	AUD 1.35	-63%	-58%
Stagflation Persists	AUD 1.12	-69%	-65%

Market reference (21 May 2026): share price AUD 3.61; consensus target AUD 3.61; EV AUD 7,682m. Anchored to H1 FY26 results (31 March 2026); valuation date 25 May 2026. WACC held constant across scenarios at 9.16% per methodology §3.5.

Key takeaways for discussion

- Asymmetry is structural. Downside / upside ratio is 4× — Stagflation drops AUD 2.10/share below Muddle Through, Orderly only adds AUD 0.50. This is intrinsic to a cyclical-industrial with elevated leverage; consensus DCFs don't usually expose it.
- Market is pricing close to Orderly Convergence. AUD 3.61 is essentially Orderly less a small haircut. Our more-sceptical Muddle Through view sits 11% below, which feels right for a central case that is more conservative than consensus.
- Single biggest firm-specific variable is the US gas-contract roll-off (2028-2030 window). Determines how much of DNL's structural cost advantage survives into the terminal state. Worth a dedicated discussion with Ben.
- Three demerger transition items remain in the equity bridge — Phosphate Hill ARO net gap (~AUD 44m), inventory supply commitment (~AUD 80m), and contingent Perdaman receipt (~AUD 73m probability-weighted).

1. World scenarios

Six scenarios were selected via the May 2026 workshop, following the §6.2 boundary-cases-plus-one-interior convention. Muddle Through is the interior most-likely; the other five are boundary cases spanning the plausible range across macroeconomic, geopolitical, climate and technology dimensions. The framework's canonical output is the comparative view across all six (per §6.2) — we explicitly do not produce a probability-weighted blended number.

Institutional triangulation

The six scenarios were sanity-checked against published institutional scenarios from the IMF World Economic Outlook, OECD Economic Outlook, World Bank Global Economic Prospects, IEA World Energy Outlook, and NGFS climate scenarios. Each of our scenarios mapped to a similar emphasis in at least one institutional set; none was idiosyncratic. Full mapping in design/scenarios_workshop.md.

The six scenarios

Muddle Through (interior, most-likely)

DM real GDP 2.0%, inflation gradually normalising to 2.5%, rates stable, no commodity boom or bust. Mining-customer capex cycle continues at current pace; demand growth tracks GDP. The market is implicitly pricing closer to Orderly Convergence than to this view.

Orderly Convergence (upside boundary)

Productivity acceleration combined with ordered geopolitics. DM real GDP 2.5%, inflation contained at 2.0%, mining real growth 4.0%, gas prices easing. Rates lower, ERP contained. Volume +3.5%, gross margin +0.5pp via better cost environment. Terminal growth +25bps.

Stagflation Persists (macro-downside boundary)

Persistent stagflation: weak real growth (0.5%), high inflation (4.5%), high gas costs (5% growth). Cost squeeze breaks pass-through (gross margin -3.5pp + input cost pass-through -3.5pp). Rates remain elevated (+200bps Rf); risk-off (+75bps ERP). Most severe downside.

Fragmentation (geopolitical-downside boundary)

Trade fragmentation and resource nationalism. Supply chains fracture; mining production growth slows; price-mix and volume both compressed (-1.5%/-1.5%). Cost pressure with partial pass-through. Country risk premium widens (+75bps) on top of higher rates and ERP.

Disorderly Climate Crystallisation (climate-downside boundary)

The scenario is about the crystallisation event, not about whether climate transition becomes disorderly (it already is). Carbon costs accelerate; mining mix shifts (thermal coal down, transition metals up); gas-led cost pressure (+6% gas). Margin -3pp; +3pp capex for decarbonisation; terminal growth -75bps as customer-base trajectory deteriorates.

AI Productivity Lag (technology-downside boundary)

AI fails to deliver economy-wide TFP gains. Modest labour-cost benefit (-0.5pp SG&A through narrow labour substitution in white-collar functions) but disappointing macro outcomes (1.7% real GDP). Mining-services industries are less exposed than tech-heavy sectors; the closest DNL gets to neutral.

2. Industry structure — Industrial Explosives

Industrial explosives is a mature, concentrated-oligopoly industry. The industry supplies bulk and packaged explosive products, electronic and non-electronic initiation systems, and on-site service offerings (blast design, surface delivery, fragmentation analysis) primarily to mining and quarrying customers. Customer demand is structurally tied to mineral-extraction volumes — open-cut mining of iron ore, coal (thermal and metallurgical), copper, gold, base metals, and increasingly transition minerals (lithium, nickel, rare earths). Demand cyclicality follows mining capex cycles rather than commodity prices directly. The industry's economics are dominated by anhydrous ammonia (and its upstream feedstock, natural gas) as the primary cost input.

Multi-tier competitive landscape

- Tier 1 — global / multi-regional. Orica (ASX, the leader), Dyno Nobel (ASX), MAXAM (Spain/Mexico, private), EPC Groupe (Euronext). Top 3 hold ~65% of global market.
- Tier 2 — large regional. Austin Powder (US private), AECI Mining Chemicals (JSE), Solar Industries (NSE), Sasol/BME (South Africa, Middle East), Yara (Norway, mining-services arm only).
- Tier 3 — country-specific / closed markets. China and Russia are effectively closed to Western majors. India is becoming Solar-Industries-led. Some country-specific incumbents in Indonesia, parts of LATAM and Africa.
- Australia is a duopoly (Orica + Dyno Nobel). North America has more competitors. Africa is a regional duopoly-to-triopoly. LATAM fragments by country.

Porter Five Forces — industry-level ratings

Force	Rating	Why
Buyer power	HIGH (binding)	Mining-customer concentration; backward-integration credibility (CSBP precedent); price-sensitive on bulk ANFO; scale leverage at contract renewal.
Supplier power	MODERATE	Ammonia and gas are regionalised. Long-term gas contracts shield producers; spot-gas exposure bears it. Interacts with buyer power on pass-through.
Threat of new entrants	LOW	Regulatory licensing; ammonia/explosives capex billions $; on-site infrastructure switching costs; expected retaliation high. Top-tier moats.
Threat of substitutes	LOW	Mechanical excavation only economic for small-scale operations. Electronic vs non-electronic initiation is intra-industry, not substitute.
Rivalry	MODERATE	Concentrated; long-term contracts dampen day-to-day competition; periodic contract-renewal events drive intensity. Geographic split also dampens direct rivalry.

Structural conclusion

Buyer power is the binding constraint. The industry's ability to pass through input-cost shocks depends on contract structure with mining customers. Under sustained input-cost inflation, mining-customer push-back blocks full pass-through — this is the mechanism that compresses margin in Stagflation and Disorderly Climate. The moderate supplier-power rating becomes more material when combined with buyer pressure on pass-through. This interaction is what drives DNL's negative asymmetry across scenarios.

3. DNL company positioning

Dyno Nobel Limited (ASX:DNL), formerly Incitec Pivot Limited, is post-demerger a pure-play industrial explosives company. The fertilisers business has been demerged; what remains is the explosives operations across North America (~55% US revenue), Australia (~35%), and a smaller rest-of-world footprint (~10% via Dyno Nobel EMEA & LATAM).

Structural advantages

- US cost-curve advantage from long-term natural gas supply contracts. Concentrated decay in the 2028–2030 window as contracts roll. Meaningful through the explicit forecast horizon. Single largest firm-specific variable.
- Long-term mining offtake contracts with the major Australian miners (BHP, Rio Tinto, FMG). 78% of revenue under multi-year contract; ~4-year weighted-average maturity. Switching-cost moat at the customer level.
- On-site manufacturing infrastructure at major customer mine sites (200+ mobile manufacturing units globally). Creates physical switching costs and embedded relationships.

Per-entity functional currency (IAS 21 / AASB 121)

DNL's functional-currency treatment is per-entity, not consolidated. Parent (Dyno Nobel Limited) is AUD-functional (Australian incorporation, ASX listing, AUD parent financing). Dyno Nobel Americas subsidiary is USD-functional (predominantly US revenue, US-denominated input cost base, US-denominated long-term gas supply contracts driving the cost advantage). Australian operations are AUD-functional. Most published sell-side AUD DCFs lose the USD-dominant operating economics of the Americas subsidiary in the FX translation noise; the framework values per-segment and translates at consolidation.

H1 FY26 result (anchor for the valuation)

- Explosives EBIT (continuing ops): AUD 224m ex IMIs, +28% pcp. EBIT margin 13.9% (vs 12.1% pcp). EBITDA margin 22.3%.
- Group NPAT ex IMIs: AUD 161m, +83%. EPS ex IMIs 9.0 cents (vs 4.7 cps).
- Net debt 31 March 2026: AUD 1,261m. Net debt / EBITDA 1.3× (well within 2.0× policy).
- FY26 EBIT guidance reaffirmed at AUD 460–500m (Explosives). FY28 ambition AUD 600m (Explosives). Implied 12% EBIT CAGR FY26–FY28, driven mostly by transformation programme margin expansion.
- Capital management: AUD 558m of AUD 900m buyback complete (AUD 342m remaining); 50% payout dividend policy; interim 4.6cps unfranked declared 11 May.
- Fertilisers separation concluded in H1: Phosphate Hill binding sale agreement (March 2026), IPF Distribution sale completed, Perdaman offtake sold to Macquarie for up to AUD 145m contingent.

Calibration anchors used in the valuation

- Base year revenue (FY26 cont. ops): AUD 3,400m. Normalised from H1 actuals + FY26 EBIT guidance, excluding discontinued Phosphate Hill.
- Base year EBIT margin (Group): 12.9% (= AUD 480m Explosives midpoint less AUD 40m Corporate, on AUD 3,400m revenue).
- Margin glide: +100/+200/+220bps Y1-Y3 then flat, capturing the AUD 300m transformation EBIT uplift management has communicated.
- Effective tax rate 22.5% (FY26 guidance midpoint, vs 30% statutory).
- WACC 9.16% (rigorous component build: Rf 4.30%, ERP 5.0%, β 1.15, after-tax Rd 4.65%, E/V 83.5%, D/V 16.5%) — held constant across scenarios.

4. DNL per-scenario impact

Per-scenario summary

Scenario	Rev. growth	Y5 margin	Term. g	Per share
Muddle Through	5.9%	15.1%	2.50%	AUD 3.22
Orderly Convergence	7.2%	15.6%	2.75%	AUD 3.72
AI Productivity Lag	5.3%	15.6%	2.25%	AUD 3.14
Fragmentation	6.1%	12.1%	2.25%	AUD 2.36
Disorderly Climate	7.3%	12.1%	1.75%	AUD 1.35
Stagflation Persists	5.2%	7.6%	2.25%	AUD 1.12

Revenue growth derived from scenario macro -> industry archetype (volume + pricing formulas) -> DNL position (90% DM weight, 10% EM, -50bps mature-market drag). Margin = base 12.9% + glide path + scenario delta. Single WACC 9.16% applied to all.

Three transmission channels that distinguish DNL

(a) Input cost pass-through under sustained inflation

The industry's moderate supplier-power rating implies that under sustained input-cost inflation, mining-customer pushback blocks full pass-through. DNL's long-term US gas contracts shield ~70% of US gas exposure through 2028; Australian gas (no long-term contract position) and ammonia spot exposure flow through. This is the binding constraint under Stagflation and also relevant under Disorderly Climate (carbon-cost flow-through has a similar mechanism). Watchable signals: Australian east-coast gas prices; US natural gas curve through 2028–2030; mining-customer contract reset patterns.

(b) Geographic / bloc-alignment exposure

DNL operates ~55% USA + ~35% Australia + ~10% RoW. Both core jurisdictions sit in the US-aligned bloc. Under Fragmentation, the US-aligned bloc is the slower-growing bloc, and DNL's cross-bloc supply-chain exposures (ammonium nitrate flows, ammonia inputs) are regulatorily sensitive. The China-aligned bloc upside doesn't accrue to DNL. Partial offset: Australia's strategic-mineral position provides demand pull from US-aligned bloc strategic-mineral supply (BHP nickel, FMG iron ore, broader copper/lithium exposure).

(c) Terminal-state coal-customer attrition

Coal-mining customers represent ~35% of DNL's customer mix (~20% thermal, ~15-20% met). Under Disorderly Climate, the moat decays faster than the standard fade-period assumes. The architecture's §10.6 rule 2 defended-exception treatment captures this without claiming a permanent breach of cost-of-capital convergence — moat decays, but converges to cost of capital over a shortened fade period. This explains the -75bps terminal-growth delta in Disorderly Climate (the only scenario where this exception fires).

5. Methodology principles applied

These were established during the 25–28 May 2026 working sessions and are documented in design/methodology/equity_bridge_and_valuation_mechanics.md (binding for the Step 6 production translator and Step 7 production DCF engine).

Five core principles

(a) Revenue growth derived from a chain, not a scalar

Scenario macro (real GDP, inflation, mining growth, gas prices) → industry archetype translation (volume + pricing formulas) → company position (geographic mix multiplier + company offset) → revenue growth path. Each layer carries explicit translation rules. Replaces the hardcoded 2% baseline used in the early smoke-test.

(b) Single WACC across scenarios

WACC is set at the valuation date and held constant across all scenarios. Each scenario already prices its risk through the cash-flow path; using a higher discount rate in a stress scenario double-counts risk. Beta already captures the systematic risk of the entire conditional cash-flow distribution. Scenario rate-driver deltas (Rf, ERP, country) are retained as narrative context but do NOT flow into the DCF discount rate. Sensitivity is via an explicit user-override cell. Terminal growth remains scenario-conditional (it represents a structural economic state, not risk re-pricing).

(c) Anchor-date discipline in the equity bridge

Every quantitative input on the company position carries an as-at date. Share count and net debt are paired to the same balance-sheet anchor (31 March 2026 for DNL H1). Period A walk-forward accounts for items between anchor date and valuation date (operating cash flow, capex, buyback, newly declared dividends). Anchor-date mismatch within a block triggers a validator warning.

(d) Structured equity bridge with on-balance-sheet flag

Each equity-bridge adjustment (declared dividend, ARO commitments, contingent receivables, restructure execution cost) carries an explicit on_balance_sheet_at_anchor flag (yes/no/partial) and a provided_for_at_anchor amount. Avoids double-counting items already in reported net debt. Contingent receivables are AASB 15 probability-weighted, not face-value.

(e) Restructure consistency rule

If the forecast assumes the benefit of a restructure (lower run-rate cost base, exit from a loss-making operation, transformation uplift), the equity bridge must include the cash cost of executing the restructure, less any existing provision. For DNL: we model the benefit of the Phosphate Hill exit and the transformation programme; we deduct the AUD ~44m ARO net gap, AUD 80m inventory commitment, and AUD 12m restructure execution cost from equity.

What this framework adds over conventional analyst views

- USD functional currency applied per IAS 21 / AASB 121 — most published Australian DCFs lose the USD operating economics in FX translation noise.
- Supplier-power-blocks-pass-through tested explicitly under sustained inflation, not assumed away.
- Coal-customer attrition reflected in terminal state via §10.6 defended exceptions — not a static terminal ROIC.
- Fragmentation modelled as a structured scenario rather than a risk-register sensitivity.

What's parked / next

- Step 6 — production translator and pydantic schema additions to make the methodology code-executable across companies. Currently the DCF runs in Excel; we want it to run from YAMLs through Python.

- Step 7 — explicit fade period for terminal-ROIC convergence (will address the 73% terminal-share-of-EV diagnostic in Muddle Through).
- Step 9 — populate CSL and WBC against the same framework. Two very different archetypes will stress-test the architecture.
- Working with Ben on: H1 vs TTM data quality issue (EODHD doesn't include half-year accounts consistently); base-year recalibration once H1 financials are curated by the data workstream.

Open methodology items (not blockers for this valuation)

- Governance / specific-risk-premium discipline — provisionally parked; Tara's preference to avoid ad-hoc alpha-style adjustments. Will revisit when populating a company where governance is materially in question.
- Beta movement under scenarios — currently treated as scenario-independent. May need company-specific scenario sensitivity if a particular scenario fundamentally changes systematic risk exposure.
- Mid-period vs end-period convention for terminal value — currently end-of-Y5. Production engine may differ.